VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
2022
HIGHER 2

NAME: ……………………………………………………………….……………

CT CLASS: …………………..………

BIOLOGY

9744 / 03

Paper 3 Long Structured and Free-response Questions

21/09/2022

2 hour

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and CT class in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the space provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in bracket [ ] at the end of each question or part question.

Question	Marks
Section A	
1	
2	
3	
Section B	
Total	

This document consists of **18** printed pages.

[Turn over]

Section A

Answer **all** the questions in this section.

1 Caffeine is a bitter, white crystalline alkaloid produced by a wide range of plants, like tea, coffee, cacao, guarana, holly, and some citrus plants.

Fig. 1.1 shows a phylogenetic tree involving some caffeine-producing plants, constructed by the comparison of the amino acid sequence of ribulose bisphosphate carboxylase (Rubisco) extracted from the plants.

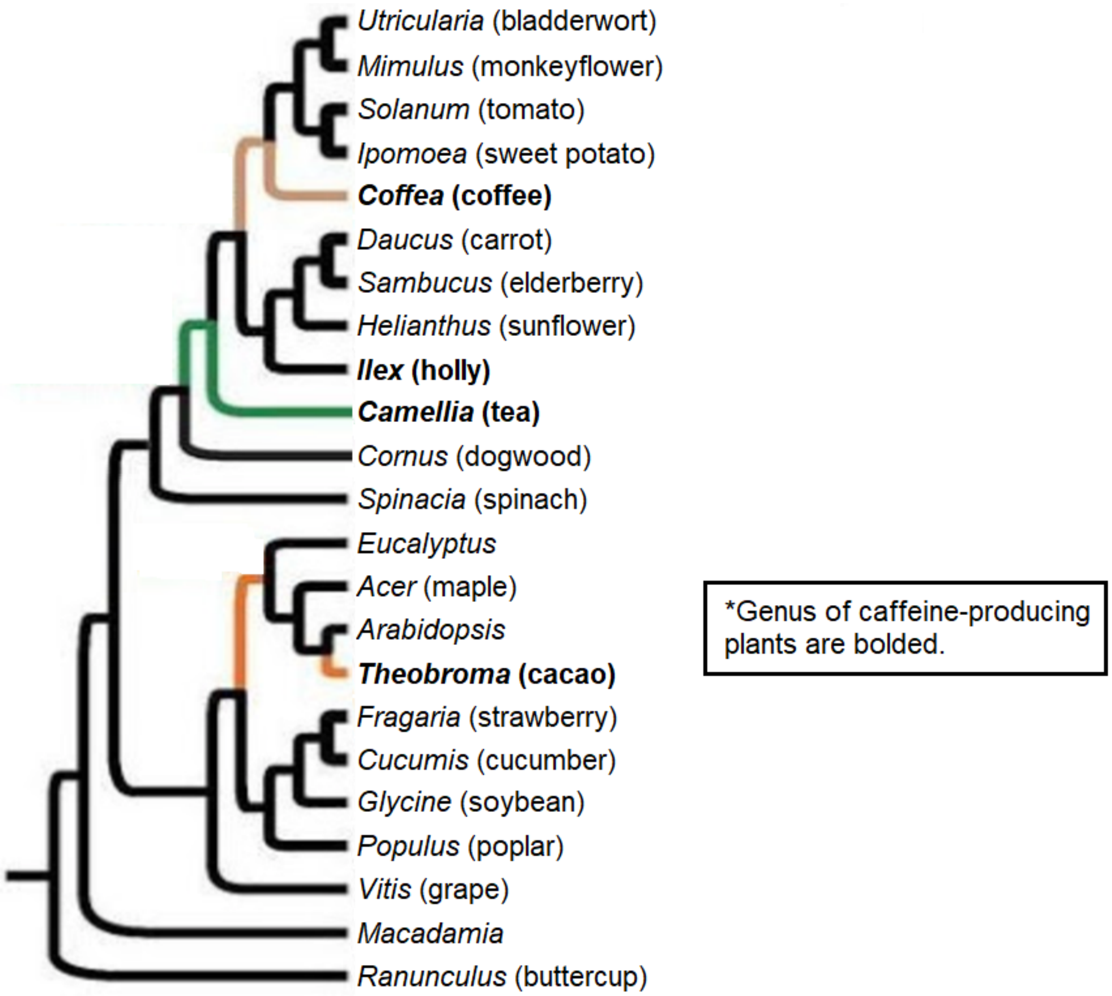

Fig. 1.1

(a) (i) State, as precisely as you can, where Rubisco is found in the plants.

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(ii) Outline the significance of Rubisco to the plants.

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With reference to Fig. 1.1,

(iii) of all the caffeine-producing plants, state the genus of the least related one.

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(iv) deduce if the evolution of caffeine in plants shows divergent evolution. Give a reason for your answer.

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(v) An evolutionary biologist commented that the use of amino acid sequence of Rubisco in the construction of the phylogenetic tree in Fig. 1.1 is reliable. However, reliability of the phylogenetic tree can be improved.

Justify the claim of the scientist that the phylogenetic tree is reliable and suggest an improvement to the reliability.

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(b) The global consumption of coffee has been increasing steadily, as shown in Fig. 1.2.

Fig. 1.3 shows the consumption of coffee in different regions around the world.

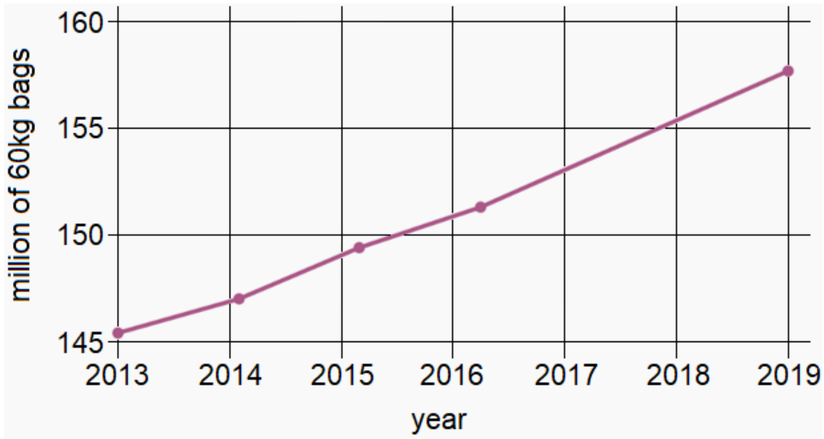

Fig. 1.2

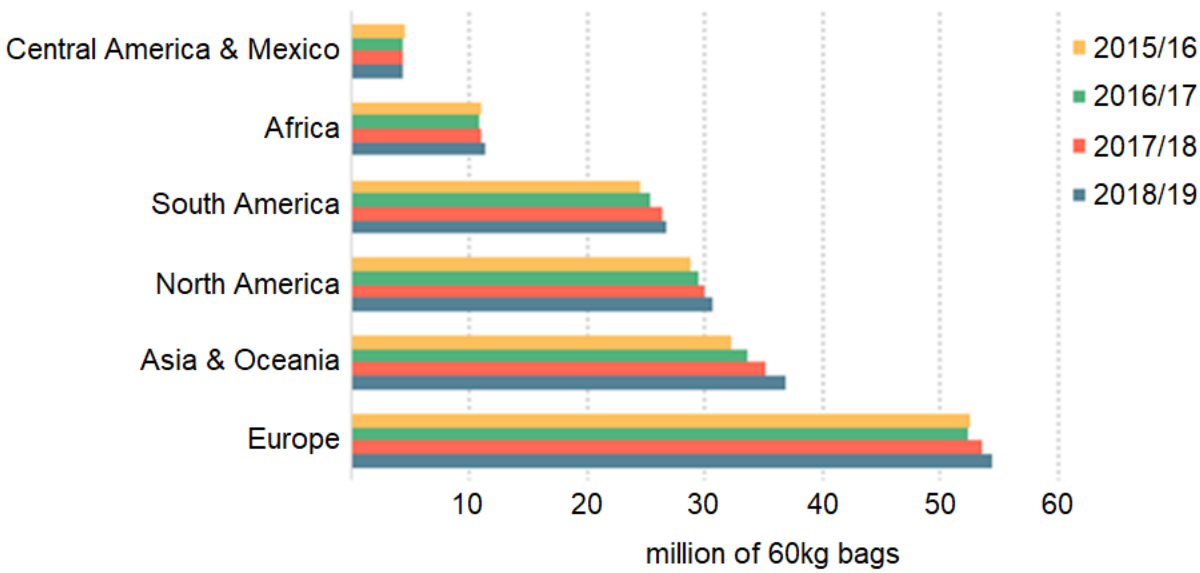

Fig. 1.3

With reference to Fig. 1.2 and 1.3,

(i) calculate the rate of increase in global consumption of coffee between 2015 and 2019. Show your working below.

[1]

(ii) state the region that contributed the most to the increase in the global consumption of coffee. Give a reason for your answer.

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(c) One reason for the increasing consumption of coffee is due to the benefits of caffeine as a cognitive enhancer, as it increases alertness and attentional performance of individuals who consume coffee.

Adenosine is a chemical found in the brain. Its concentration has been shown to increase when we are awake and active during the day. It binds to specific receptors on neurones, which causes the feeling of drowsiness. The number of receptors bound by adenosine increases throughout the day, resulting in increasing feeling of drowsiness.

Fig. 1.4 shows the signalling pathway of adenosine, and Fig. 1.5 shows the structure of adenosine and caffeine.

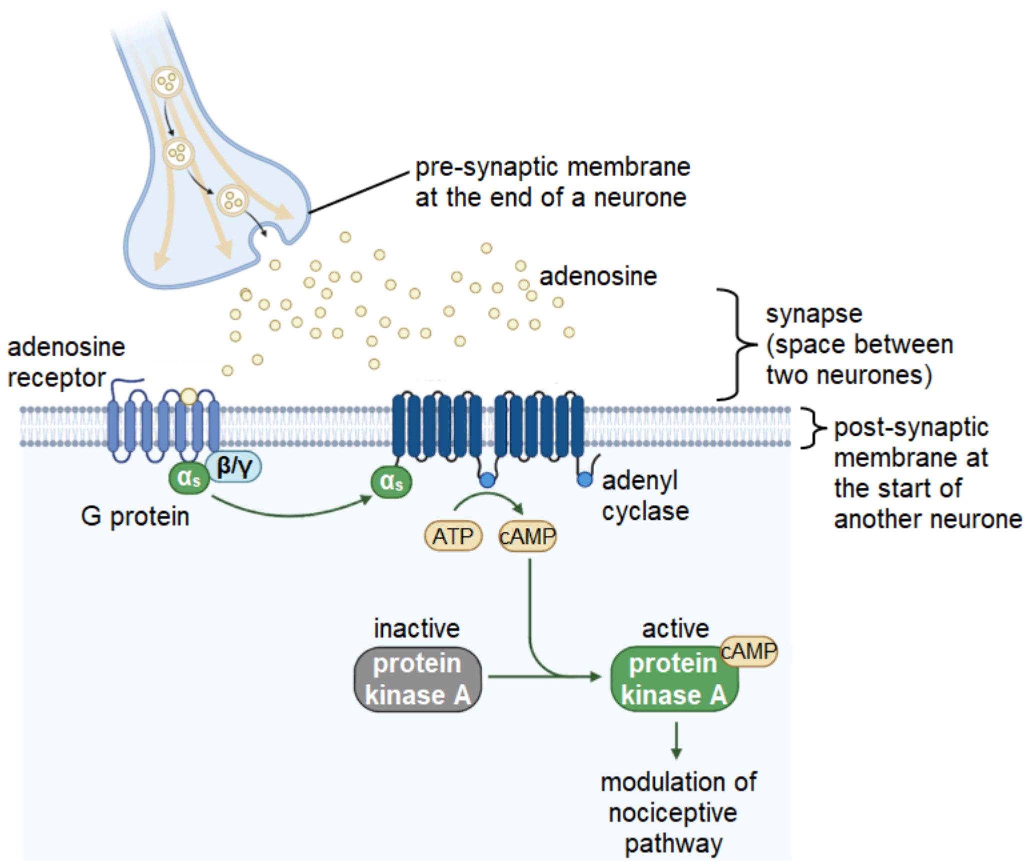

Fig. 1.4

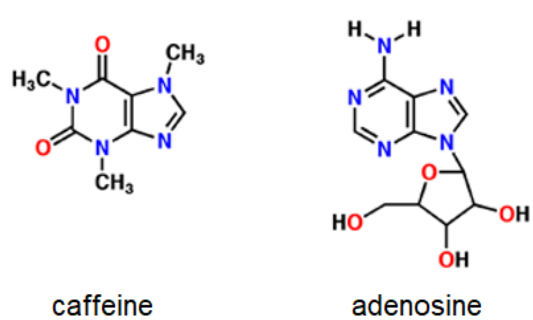

Fig. 1.5

With reference to the information provided,

(i) explain how the structure of adenosine aids in its movement across the synapse.

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(ii) explain the significance of cAMP in adenosine signalling.

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(iii) explain how caffeine functions as a cognitive enhancer.

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(iv) suggest two reasons for the need of repeated consumption of coffee throughout the day in order to prevent feeling of drowsiness.

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(d) While caffeine serves as a cognitive enhancer in humans, it serves important functions in plants. In high concentrations, it plays a pesticidal role against many insects, snails, slugs, even some fungi and bacteria. However, the presence of low concentrations of caffeine in nectar increases the chance of pollination as pollinators like bees have been observed to preferentially visit the flowers of caffeine-producing plants.

However, a species of beetle known as coffee berry borer, is able to consume the fruits and leaves of coffee plants despite the high concentration of caffeine. The resistance to caffeine in the beetle is due to the symbiotic relationship with a species of bacteria in its gut, which is able to utilise caffeine as a food source.

(i) Outline the symbiotic relationship between coffee berry borer and its gut bacteria.

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(ii) Explain how coffee berry borer may be classified as a distinct species.

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[Total: 27]

2 The cancer stem cell (CSC) theory, where scientists believe that in every tumour, a small group of cancer cells exhibit stem cell-like behaviours and properties, is gaining increasing attention from researchers and has become an important focus of cancer research.

(a) (i) Outline how cancer cells can be 'stem cell-like'.

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Under natural circumstances in the body, another group of cells, the germ line cells, also exhibit stem cell-like properties. However, they are not the same as cancer cells.

(ii) Describe the differences that can be seen in cancer cells compared with normal germ line cells.

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(b) A bone marrow transplant is also known as a stem cell transplant. Transplantation can be used to treat certain types of cancer, such as leukaemia, myeloma, and lymphoma, as well as other blood and immune system diseases that affect the bone marrow. In medical procedures, stem cells for an **autologous transplant** come from the patient's own body.

Sometimes, cancer is treated with high-dose, intensive chemotherapy or radiation therapy treatment.

(i) Suggest a reason why doctors remove the patient's stem cells from their blood or bone marrow before the cancer treatment begins.

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In blood cancer treatment, bone marrow transplant often yielded positive outcomes in many clinical trials, with over 50% of the patients surviving without remission.

A team of doctors decided to incorporate this treatment with high doses of chemotherapy in a new clinical trial. Fig 2.1 shows the Kaplan-Meier Estimates of the overall survival of patients with metastatic breast cancer, who were randomly assigned to treatment with either conventional-dose chemotherapy alone, or with high-dose chemotherapy plus autologous haematopoietic stem-cell transplantation.

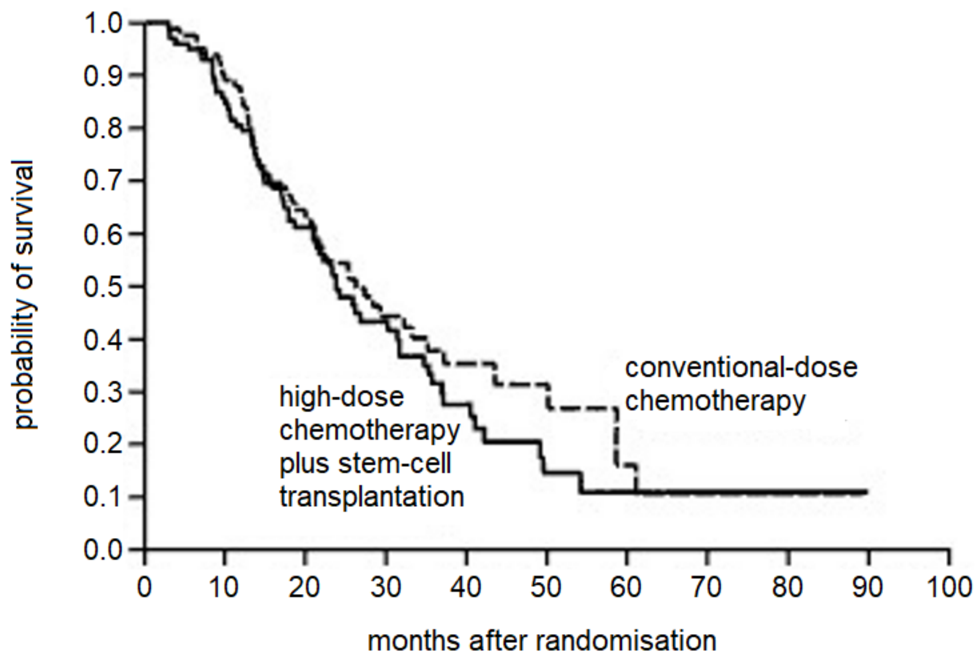

Fig. 2.1

(ii) With reference to Fig 2.1, give a conclusion for the effectiveness of the two treatment methods for breast cancer in women. Justify your answer.

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[Total: 9]

3 Coronavirus disease (COVID-19) is an infectious disease caused by the SARS-CoV-2 virus. Most people infected with the virus will experience mild to moderate respiratory illnesses and recover without requiring special treatment. However, some individuals will become seriously ill and require medical attention. Older people and those with underlying medical conditions like cardiovascular diseases, diabetes, chronic respiratory diseases, or cancer are more likely to develop serious illnesses. Anyone can get infected with COVID-19 and become seriously ill or die at any age.

The understanding of the structure of the SARS-CoV-2 virus is important for the production of drugs and vaccines.

Fig. 3.1 shows the general structure of the SARS-CoV-2 virus.

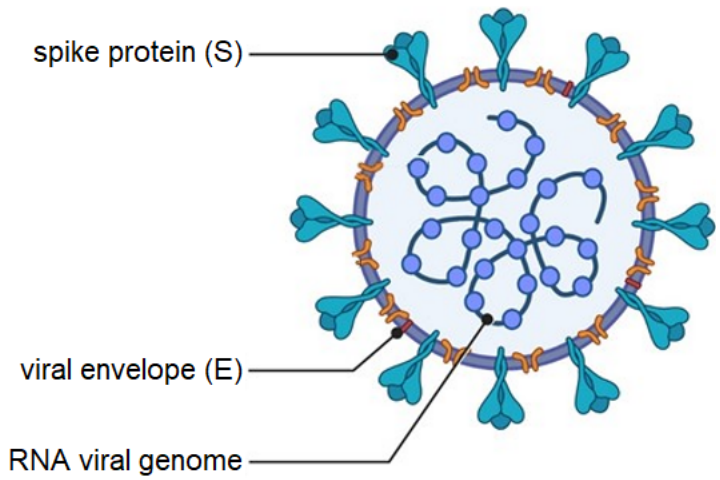

Fig. 3.1

(a) With reference to Fig. 3.1, contrast the structure of SARS-CoV-2 with that of HIV.

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(b) Researchers looked at COVID-19 patients and found that they had much higher levels of antibodies against SARS-CoV-2 than those without the disease.

Describe three ways in which the structure of antibodies contributes to their functions.

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(c) Fig. 3.2 shows the structure of immunoglobulin genes which code for antibodies.

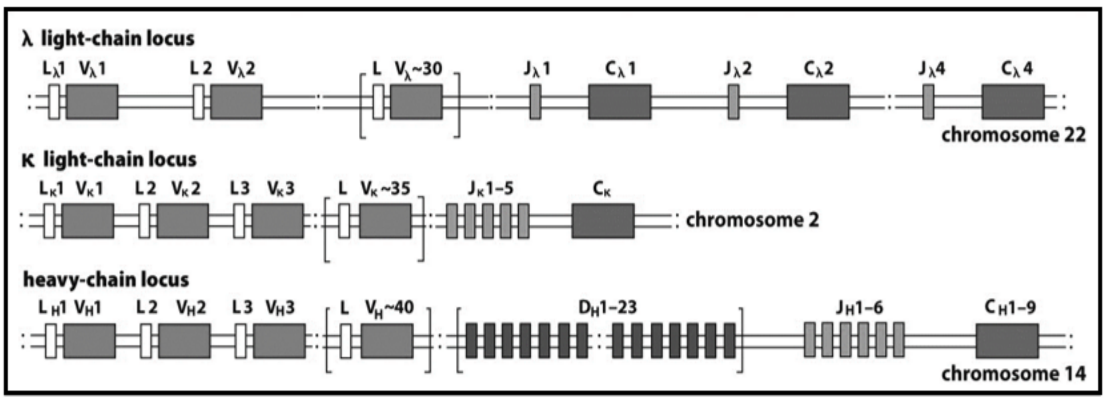

Fig. 3.2

With reference to Fig. 3.2, explain how variability at the DNA level result in variability in the antigen-binding sites of antibodies.

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(d) mRNA COVID-19 vaccines are currently being administered as a measure to reduce the severity of illnesses brought about by the infection. The vaccine contains an mRNA which codes for a spike protein. When the vaccine is injected into the upper arm muscle, the mRNA enters a cell and gets translated into a spike protein. The protein is then displayed on the cell surface membrane and gets recognised as 'foreign' by the immune cells. This triggers the production of antibodies against the spike protein.

Explain how the vaccine triggers the production of antibodies and how the antibodies confer protection against the virus.

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(e) While COVID-19 vaccines took several months to be developed, no effective HIV vaccine has been developed after almost 40 years. Scientists that are involved in the research and development of the HIV vaccine say that the problem lies not in the lack of funding but in the HIV virus itself.

Suggest why scientists have not been able to create an effective vaccine against HIV.

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.... [1]

[Total: 14]

Section B

Answer **one** question in this section.

Write your answers in the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 **(a)** Epigenetics is the study of changes in organisms that are caused by modifications of gene expression rather than alteration of the genetic code. Describe how such modifications can be achieved. [15]

(b) Using relevant examples, discuss the significance of regulating gene expression. [10]

[Total: 25]

5 **(a)** Respiration produces ATP. Describe the role of ATP in living organisms. [15]

(b) Using relevant examples, discuss the significance of complementarity in the production of ATP. [10]

[Total: 25]

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